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A Bayesian state-space model for estimating pup production in an asynchronous breeder: the grey seal (*Halichoerus grypus*)

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For wide-ranging or cryptic species, abundance estimation often relies on counts when individuals aggregate during breeding or migration. When these events are asynchronous, counts represent an unknown proportion of the focal component of the population. Grey seals (*Halichoerus grypus*), colonial breeders with protracted pupping seasons, typify this challenge: a single survey corresponds to an unknown proportion of total pup production (number of pups born in a season). Pups are born with white pelage before moulting and leaving, allowing a time series of counts of white and moulted pups to be derived from serial aerial photography. Current maximum likelihood approaches have limited flexibility to incorporate model extensions or test alternative parameterizations. We developed a Bayesian state-space model that explicitly incorporates mortality and examines effects of misspecifying moult and leave timing. Via simulation, we demonstrate biases from omitting mortality, misspecifying leave timing, and misparametrizing the observation and process models. Incorporating platform-specific (drone and fixed-wing surveys) observation error from Farne Island colonies surveyed in 2023, we refine detection rates and fit the model to these data. This framework has potential to improve abundance estimates for other

1. Introduction

Abundance estimates are critical in ecology to provide information on population size and trends that can inform conservation and management efforts. Often, this estimation is challenging because not every individual is available to be counted within a population or subpopulation. Many different methods are used, each of which has advantages and limitations depending on the species and the ecosystem being studied. For example, distance sampling (e.g. line transect surveys) is widely used for inconspicuous or wide-ranging species [1], and capture–recapture methods are applied for populations where individuals can be identified or marked [2]. For species that congregate during part of their annual cycle (e.g. at migration stopover points or breeding colonies; [3,4]), counts of these aggregations can potentially be used as an efficient means to estimate abundance.

The utility of counts from single surveys of congregations depends on the duration and synchronization of the annual cycle event. In some species that congregate to breed (e.g. some species of seabirds, amphibians, pinnipeds), at some point in the season a single survey count of the focal species or their nests provides a useful index of the population (e.g. breeding pairs; [5]). In such species, the duration of the breeding season for any one individual and the level of synchronization means that almost all breeders are present at any one time, or indicators of breeding (e.g. apparently occupied puffin burrows) can be used. For species for which the congregation is protracted or where time spent at the location of congregation is relatively short, count data resulting from single surveys may not be a reliable index of abundance; the stay of an individual at the location may be shorter than the overall event duration [6–8]. In cases such as these, the number of individuals detected often represents a variable proportion of a component of the population present across time and space [9].

Grey seals (*Halichoerus grypus*) offer a case study to investigate the estimation of population indices when a single survey does not represent the index as a whole. Grey seals inhabit the North Atlantic coast and the Baltic Sea. They are generally wide-ranging seals, spending prolonged periods at sea. During the breeding season, females typically gather in large groups (colonies), and give birth to a single pup on rocks, grass, sandy beaches or ice. However, the breeding season is more protracted than the stay of any individual pup. Grey seal pups are born with white coats (lanugo), and then moult into adult-like pelage before they leave for sea [10–13]. Pups can be categorized into five classes related to age: classes 1–3 represent whitecoat pups that have not started moulting their lanugo coat, 4 (which starts around 16 days of age) represents the transition into moulting, and 5 (which starts around 23 days of age) represents a moulted pup. A pup is considered fully moulted (i.e. in class 5) when less than 5% of the lanugo coat remains. Estimates for the total time a pup spends on the colony before going to sea vary but in the UK the estimated mean is 31.5 days [10].

The total number of births, referred to as pup production, is estimated across the grey seal range with varying methods. In some areas with small colonies, individual pups are tracked through the season. In many colonies, this is not tractable, and multiple pup count surveys are conducted. Depending on the type of survey, pup production is then estimated using a statistical model or the highest observed count is used directly. The majority of colonies in the UK are monitored by the Sea Mammal Research Unit (SMRU) at the University of St Andrews, which conducts multiple (4–5) aerial photographic count surveys throughout a season and classifies detected pups as either whitecoat (classes 1–4), moulted (class 5) or dead (almost all of which are whitecoats). Pup production estimates are derived from pup counts arising from these surveys and a model is fitted using maximum likelihood [14].

The maximum likelihood model relies on fixed stage transition and observation parameters derived from a mixture of expert opinion and earlier published and unpublished studies [10,14] (see table 2). It does not permit flexible estimation or sharing of information across colonies and years, and it does not explicitly account for mortality: dead pups are added to the count of alive whitecoats. Accurately reflecting grey seal pup ontogeny in a mechanistic framework is challenging because some parts of the life history are unobservable (e.g. leaving the colony for sea, or dying and subsequently washing out to sea) and transition timings (e.g. moulting and leaving for sea) may be misspecified. In addition, the observation-process errors that arise from (i) missed detections and (ii) misclassification between whitecoat and moulted classes when processing photographs from aerial surveys, rely on limited data. Recent

upgrades to SMRU's digital camera system, together with paired unmanned aerial vehicle (UAV) and fixed-wing surveys, provides empirical information to refine the observation parameters [15].

To address these limitations, we first approximated the maximum-likelihood pup production model described by Russell *et al.* [14] and adapted by Jacobson *et al.* [16] in a Bayesian state-space framework [17,18], mirroring their biological and observation processes to enable direct comparison (table 2; objective 1). We then extended the framework to explicitly include pup mortality and carcass persistence, and evaluated performance with simulated and real survey data (objective 2). Next, we conducted sensitivity analyses targeting (i) the observation process (detection and misclassification), (ii) transition processes (moulting and leaving) and (iii) potential confounding between detection and transitions (objective 3). Guided by these results, we reparameterized the observation process using empirically derived detection and classification rates from newly available tandem UAV and fixed-wing aerial survey data (see survey platforms in §2 and Russell *et al.* [15]) and fitted the extended model to relevant colonies (objective 4). Finally, we relaxed assumptions on moulting and leaving so these transition rates could be estimated directly from the data (objective 5). The flexibility afforded by this Bayesian state-space approach will also pave the way for other applications to species with similarly asynchronous congregations.

2. Methods

All Bayesian models considered in this study rely on a deterministic birth and transition process and stochastic observation process. Below, we describe the full model, but note that the model used in the first objective (mirroring the Russell *et al.* [14] and Jacobson *et al.* [16] maximum-likelihood models) omits the mortality and carcass persistence parameters. A table of notation and corresponding definitions used in this study can be found in supplementary material, appendix S1, table S1.1.

2.1. Full model formulation

We developed a process model for births and pup development during the breeding season. For notational simplicity we focus on a single colony and year, although the model is readily generalized to multiple colonies and years (with parameters potentially shared across these or treated hierarchically with random effects of colony and year, see §4).

The breeding season in this model begins on 1st September (day 1 of the breeding season) and ends on 1st January (day 124 of the breeding season). For modelling purposes, we divided the season into $s = 1, \dots, S$ consecutive, disjoint equal-length intervals, each τ days long. In practice, we used 4-day intervals, so $\tau = 4$, $S = 31$, and $s = 1$ means days 1–4, $s = 2$ means days 5–8, etc.

2.1.1. Birth cohorts

Let N be the total number of births across the breeding season at a given colony and year. The set of pups born in a single time interval is referred to as a 'cohort'. Cohorts are indexed by the interval, denoted c , and the number of pups born in interval c is denoted $b(c)$. We assumed no pups can be born after day 100 of the breeding season, so the maximum value of c is $C = 25$ and $\sum_{c=1}^C b(c) = N$. Births outside this window are very rare given the estimated birth distribution parameters. For the colonies and season considered here, less than 0.1% of the skew-normal distribution falls beyond day 100.

We modelled the number of births in each cohort as a deterministic process:

$$b(c) = Np_b(c),$$

where $p_b(c)$ is the probability that a pup born at some point during the breeding season is born into cohort c . We refer to $p_b(c)$ as the 'birth phenology process'. This process is controlled by a skew-normal distribution:

$$p_b(c) = \frac{\phi(d(c); \mu, \sigma, \delta)}{\sum_{\bar{c}=1}^C \phi(d(\bar{c}); \mu, \sigma, \delta)}, \quad (2.1)$$

where $d(c)$ is a function that returns a day representing the mid-point of the cohort c (given by $d(c) = \lfloor \tau(c - 0.5) \rfloor$), and $\phi(x; \mu, \sigma, \delta)$ is the probability density function of a skew-normal distribution [19] with mean μ , standard deviation σ and skewness δ evaluated at x . (Note that the skew-normal distribution is

often parametrized in terms of location, scale and shape and we provide the conversion in supplementary material, appendix S1.) The denominator in equation (2.1) ensures $\sum_{c=1}^C p_b(c) = 1$. The birth probability for each interval could alternatively be computed as the difference between the skew-normal cumulative distribution function (CDF) evaluated at the interval boundaries. Given the coarse temporal resolution of survey data and negligible differences between approaches, we retain the midpoint formulation for computational simplicity (see supplementary material, appendix S3).

2.1.2. Transition through life stages

We track each cohort through time. Let $\mathbf{x}_c(a)$ be a vector containing the number of pups in each stage for cohort c at age a (in units of τ), where stages are: (i) whitecoats, (ii) moulted pups, (iii) pups that have left the colony alive, (iv) dead pups visible on the colony and (v) dead pups that are washed away or no longer available to be counted. For a cohort born in interval c , their age in interval s (assuming $s \geq a$) is $a = s - c + 1$. All animals are born as whitecoats, so $\mathbf{x}_c(1) = [b(c), 0, 0, 0, 0]'$. Thereafter, the number in each stage is a deterministic function of age, where transitions between states are controlled by an age-specific transition matrix $\mathbf{T}(a)$ such that

$$\mathbf{x}_c(a+1) = \mathbf{T}(a)\mathbf{x}_c(a).$$

The transition matrix is given by

$$\mathbf{T}(a) = \begin{bmatrix} \rho^{[\tau]}(a)(1 - \gamma^{[\tau]}(a))(1 - \zeta^{[\tau]}(a)) & 0 & 0 & 0 & 0 \\ \rho^{[\tau]}(a)\gamma^{[\tau]}(a)(1 - \zeta^{[\tau]}(a)) & \rho^{[\tau]}(a)(1 - \zeta^{[\tau]}(a)) & 0 & 0 & 0 \\ \rho^{[\tau]}(a)\zeta^{[\tau]}(a) & \rho^{[\tau]}(a)\zeta^{[\tau]}(a) & 1 & 0 & 0 \\ (1 - \rho^{[\tau]}(a)) & (1 - \rho^{[\tau]}(a)) & 0 & \chi^{[\tau]} & 0 \\ 0 & 0 & 0 & 1 - \chi^{[\tau]} & 1 \end{bmatrix},$$

where survival ($\rho^{[\tau]}(a)$), moulting ($\gamma^{[\tau]}(a)$), leaving ($\zeta^{[\tau]}(a)$) and carcass persistence ($\chi^{[\tau]}$) are described below. The superscript $[\tau]$ indicates that transition probabilities are calculated over interval widths. We allow for the small probability of a whitecoat leaving the colony alive, reflecting that a pup can be physiologically moulted while still being correctly classified as a late-stage whitecoat (i.e. minimal white lanugo remaining), and may leave the colony within the same time interval as completing the moult.

Survival. Age-specific survival probability is assumed to increase linearly on the logit scale with age and is modelled as $\rho^{[\tau]}(a) = (\rho^{[d]}(a))^{\tau}$, where $\rho^{[d]}(a)$ is daily survival probability for a pup in age interval a and is given by $\rho^{[d]}(a) = \text{logit}^{-1}(\beta_0 + \beta_1(a-1)\tau)$. Note we anticipate that $\beta_1 > 0$ (to give an increase in survival with age), and that $a\tau$ converts age in intervals to age in days.

Moulting. We assumed the probability of a whitecoat pup becoming fully moulted ($\gamma^{[\tau]}(a)$) at age a is controlled by a cumulative Gaussian distribution:

$$\gamma^{[\tau]}(a) = \begin{cases} \Phi(\tau; \mu_m, \sigma_m), & a = 1, \\ \frac{\Phi(a\tau; \mu_m, \sigma_m) - \Phi((a-1)\tau; \mu_m, \sigma_m)}{1 - \Phi((a-1)\tau; \mu_m, \sigma_m)}, & a > 1, \end{cases}$$

where $\Phi(x; \mu, \sigma)$ is the CDF of a Gaussian distribution with mean μ and standard deviation σ evaluated at x . In the $a > 1$ case, the numerator of the fraction is the probability of transition at age a and the denominator the probability a pup has not transitioned by age $a-1$. Hence, $\gamma^{[\tau]}(a)$ is the probability a whitecoat pup moults in interval a conditional on it not having moulted in a previous interval. To avoid potential computational issues when the denominator approached zero, we set $\gamma^{[\tau]}(a) = 1$ when $\Phi((a-1)\tau; \mu, \sigma) > 1 - 10^{-6}$. Transitions are evaluated over discrete intervals of length τ ; however, the CDFs are defined in continuous time (days). Evaluating Φ at $a\tau$ allows us to link the continuous-age specific transition processes to discrete-time transition probabilities used in the transition matrix $\mathbf{T}(a)$. Given strong prior information, we fixed the moulting process parameters at $\mu_m = 23$ days and $\sigma_m = 5$ days for objectives 1–4. We relaxed this assumption in objective 5 and let the model estimate moulting parameters.

Leaving the colony. Similarly to moulting, the probability of a pup leaving the colony ($\zeta^{[r]}(a)$) during interval a is

$$\zeta^{[r]}(a) = \begin{cases} \Phi(\tau; \mu_l, \sigma_l), & a = 1, \\ \frac{\Phi(a\tau; \mu_l, \sigma_l) - \Phi((a-1)\tau; \mu_l, \sigma_l)}{1 - \Phi((a-1)\tau; \mu_l, \sigma_l)}, & a > 1. \end{cases}$$

To avoid computational issues, we set $\zeta^{[r]}(a) = 1$ when $\Phi((a-1)\tau; \mu, \sigma) > 1 - 10^{-6}$. To mirror previous studies (objective 1), we initially fixed the leaving process parameters at $\mu_l = 31.5$ days and $\sigma_l = 7$ days. We kept these fixed values for objectives 2–4. However, we note that there is currently limited information about the age at which pups leave UK colonies and the variation around this mean. For objective 5, we therefore allowed the model to estimate these parameters.

Carcass persistence. Carcass persistence was assumed constant across age, with $\text{logit}(\chi^{[d]}) = 4$, implying a daily persistence probability $\chi^{[d]} = 0.982$. Equivalently, approximately 70% of carcasses remain visible after 20 days since $(\chi^{[d]})^{20} \approx 0.70$. The persistence probability over interval width τ is $\chi^{[\tau]} = (\chi^{[d]})^\tau$. Empirical data were unavailable for dead pups, so we chose a value that represents a plausible decay rate, where carcasses remain available to be detected for a few weeks before they become unavailable owing to decay, scavenging or washing out to sea. We also examined shorter persistence scenarios in simulation (e.g. $\text{logit}(\chi^{[d]}) = 2.5$ or 3.5 (corresponding to $\chi^{[d]} = 0.924$ or 0.971 , respectively)).

Given the above, the total number of pups in each state at time interval s is

$$\mathbf{n}(s) = \sum_{c=1}^{\min(s,C)} \mathbf{x}_c(s-c+1),$$

where $\mathbf{n}(s)$ is a vector of equal dimension to $\mathbf{x}_c(a)$.

2.1.3. Observation model

The model includes an observation process linking the number of pups in each state during time intervals where there are observations with the observed counts. We allow for pups to be missed and for misclassification of moulted pups as whitecoats. Let $\mathbf{o}(s) = [o_w(s), o_m(s), o_d(s)]'$ be a vector containing the expected count of whitecoat, moulted and dead pups in an interval s within which an aerial survey took place. This count is controlled by an observation process matrix, $\mathbf{O}$, such that

$$\mathbf{o}(s) = \mathbf{O}\mathbf{n}(s).$$

The observation process matrix is given by

$$\mathbf{O} = \begin{bmatrix} \psi_w & \psi_m(1-\kappa) & 0 & 0 & 0 \\ 0 & \psi_m\kappa & 0 & 0 & 0 \\ 0 & 0 & 0 & \psi_d & 0 \end{bmatrix},$$

where ψ_w is the probability of observing a white pup, ψ_m is the probability of observing a moulted pup, κ is the probability of correctly classifying a moulted pup as such and ψ_d is the probability of observing a dead pup.

The observed counts of each stage are assumed to follow independent Poisson distributions:

$$y_w(s) \sim \text{Poisson}(o_w(s)),$$

$$y_m(s) \sim \text{Poisson}(o_m(s)),$$

$$y_d(s) \sim \text{Poisson}(o_d(s)),$$

where $y_w(s)$, $y_m(s)$ and $y_d(s)$ are the, respective, observed counts for white, moulted and dead individuals in time interval s . Our objective 1 was to mirror previous likelihood-based models [14,16], and in those models, there was no separate accounting for dead pups; instead, observed dead pups were added to the count of whitecoats. We replicated this by simplifying the expected count vector to $\mathbf{o}(s) = [o_w(s), o_m(s)]'$, amending the observation process equation to

Table 1. Model parameters and prior distributions. Lognormal priors are used for positive location parameters (e.g. total production, mean birth date and mean leave age), gamma priors for scale parameters and a normal prior for the skewness parameter truncated at zero (equivalently, a half-normal prior) to enforce right-skewness.

parameter	description	prior distribution (natural scale)
N	total pup production	lognormal(5, 2)
μ_b	mean birth date	lognormal(3.5, 0.15)
σ_b	s.d. of mean birth date (days)	gamma($k = 2$, $\theta = 6$)
α_b	birth date skewness (days)	normal(0, 6)
μ_m	mean age at moult (days)	normal(23, 2)
σ_m	s.d. of age at moult (days)	gamma($k = 2$, $\theta = 4$)
μ_l	mean age at leaving colony (days)	lognormal(3.5, 0.1)
σ_l	s.d. of age leaving colony (days)	gamma($k = 1$, $\theta = 5$)

$$\mathbf{O} = \begin{bmatrix} \psi_w & \psi_m(1 - \kappa) & 0 & \psi_d & 0 \\ 0 & \psi_m\kappa & 0 & 0 & 0 \end{bmatrix}$$

and modelling observed counts as

$$y_w(s) + y_d(s) \sim \text{Poisson}(o_w(s)),$$

$$y_m(s) \sim \text{Poisson}(o_m(s)).$$

2.2. Prior specification

For objectives 1 and 2 (mirroring and extending the standard maximum likelihood model, respectively), we specified weakly informative priors for total pup production and birth date mean, standard deviation and skew. These priors constrained the model to biologically plausible ranges while maintaining flexibility (see supplementary material, appendix S3). Parameters that are constrained to be strictly positive (e.g. total pup production, birth process and transition processes) were modelled on logarithmic scales, while other parameters were re-scaled to avoid values too close to zero. We also used a non-centred parameterization to improve convergence and avoid funnel-shaped posterior distributions [20]. Prior specifications are listed in table 1. For objective 3 (sensitivity analyses), we used the same weakly informative priors as in objectives 1 and 2 (table 1). In addition to estimation, time to leave and moulting parameters were additionally fixed or varied systematically to reflect different scenarios. For objective 4 (reparameterized full model), we again used the same weakly informative priors as in objectives 1 and 2, but incorporated the updated observation parameters (see supplementary material, appendix S2). For objective 5 (estimating moulting and leaving times), we introduced moderately informative priors for these transition parameters to restrict estimates to biologically plausible ranges while allowing flexibility (summarized in table 1).

2.3. Validation of interval births and observation process

The model formulates birth timing in discrete intervals. To validate the interval approach, we evaluated whether simulating daily births and modelling interval births introduced bias into pup production estimates ($n = 30$). This simulation determined if evaluating skew-normal births at interval midpoints could capture the continuous birth process.

Next, we compared the Poisson and the multivariate normal (MVN) approximation to the multinomial observation likelihoods using simulated datasets ($n = 30$). The MVN likelihood treats observed counts as approximately normally distributed and adds a covariance matrix to reflect dependence

between stage counts inherent in the multinomial sampling process. The multinomial was not used because imperfect detection and misclassification mean the observed counts do not satisfy the fixed total N assumption of the multinomial likelihood.

Results of these validation analyses are provided in supplementary material, appendix S3.

2.4. Model implementation

All Bayesian state-space models (across objectives) were implemented in R version 4.4.3 [21] using the RStan package [20]. Models were fitted with Hamiltonian Monte Carlo sampling. For each model, we ran four Monte Carlo Markov chains (MCMC) with 2000 iterations each, discarding the first 1000 iterations per chain as warm-up. Convergence was assessed using the Gelman–Rubin diagnostic [22] and visual inspection of trace plots. Uncertainty was assessed using the HDInterval package in R [23]. The 95% credible interval (CRI) coverage rate is defined as the proportion of simulations for which the true parameter value fell within the 95% highest density CRI of the posterior distribution. Goodness of fit was evaluated through posterior predictive checks and Bayesian posterior predictive p -values for observations, with values close to 0.5 indicating a good model fit [22].

2.5. Analyses

We fitted the model to both simulated and real datasets. Simulated data allowed us to quantify bias and coverage of parameter estimates. Real datasets from Brownsman and Staple Island in the Farne Islands (UK) allowed us to assess how well the model fitted real data and compare results across different survey platforms and observation parameters. Below, we describe the specific analysis for each objective.

For simulations, we simulated a total pup production of $N = 500$ and simulated five surveys spaced 12 days apart to represent a realistic survey scenario [14]. Empirical analyses were based on a subset of the Farne Island colonies in 2023.

Objective 1: Mirror standard model in Bayesian framework. We mirrored the maximum-likelihood model approach of Russell *et al.* [14] and Jacobson *et al.* [16] in a Bayesian framework using previously published fixed detection and classification rates ($\psi_w = 0.95$, $\psi_m = 0.95$, $\kappa_m = 0.91$, $\kappa_w = 1.0$; table 2). We fitted this model to simulated data where mortality was present but not explicitly modelled (i.e. dead pups were added to whitecoat counts as in these previous approaches).

Objective 2: Extend model to explicitly include mortality. We extended the framework to explicitly model pup mortality and carcass persistence. To evaluate performance, we simulated datasets with plausible mortality rates of grey seal colonies: daily survival probability of approximately 0.98 for newborn pups (1.8% daily mortality), increasing linearly with age. We then fitted the full model (with mortality explicitly modelled) and compared results to the standard model (objective 1, which omitted mortality). We also fitted the model to real survey data from Brownsman and Staple Island using the observation parameters from table 2.

Objective 3: Conduct sensitivity analyses. We systematically varied observation parameters (detection and misclassification rates) and transition process parameters (moulting and leaving) around their assumed values to test robustness of pup production estimates. We also tested assumptions about estimation of time to moult (TTM) and time to leave (TTL). For observation parameters, we varied ψ_w , ψ_m , κ_m and κ_w across plausible ranges (0.65–1.0; see supplementary material, appendix S4) to assess sensitivity to detection and classification errors. For transition processes, we tested TTM values ranging from 20 to 26 days and TTL values from 20 to 40 days to evaluate sensitivity to misspecification of these parameters. We also examined potential confounding between TTL and κ_m and the effect on pup production by estimating the leaving process (mean and standard deviation) at a range of κ_m values. For each scenario, we simulated datasets ($n = 50$ replicate simulations with 500 total births). Detailed methods and complete results of sensitivity analyses are provided in supplementary material, appendix S4.

Objective 4: Survey platforms and updated observation parameters. In 2023, a rare tandem UAV and fixed-wing survey was conducted on a single day (28 November 2023), allowing for a comparison of detection and classification rates for white and moulted pups across survey platforms (see Russell *et al.* [15] for detailed description of survey platforms and supplementary material, appendix S2, table S2.3, for rates used to fit the model to Brownsman and Staple Island data).

To evaluate how different survey platforms influenced detection and classification (objective 4), we used paired UAV and fixed-wing surveys at Brownsman and Staple Island. UAV flights were conducted

Table 2. Values used to simulate biological and observation processes (objectives 1 and 2). Descriptions of each fixed parameter are provided for clarity, and units are given in parentheses where applicable.

biological process		
parameter	description	value
D	length of breeding season (days)	123
μ_b	mean birth date (days)	34
σ_b	s.d. of birth date (days)	12
μ_m	mean age at moult (days)	23
σ_m	s.d. of age at moult (days)	5
μ_l	mean age at leaving colony (days)	31.5
σ_l	s.d. of age at leaving colony (days)	7
observation process		
ψ_w	detection probability of whitecoat pups	0.95
ψ_m	detection probability of moulted pups	0.95
κ_m	probability of correctly classifying moulted pups	0.91
κ_w	probability of correctly classifying whitecoat pups	1.0

at both low (47 m) and high (65 m) altitudes and aligned in time with fixed-wing aerial surveys. Low-altitude UAV flights (47 m) were assumed truth. High UAV surveys were conducted on the same 11 dates between 4th November and 18th December, with 3–8 days between surveys, and non-consecutive low UAV surveys were conducted on six of those dates. The UAV surveys were conducted by Skeye Ltd, commissioned by National Trust. SMRU fixed-wing aerial surveys, which represent the lower-resolution photographic survey method, were of a similar temporal extent but lower temporal resolution; five surveys were conducted between 31st October and 13th December, spaced 7–15 days apart. Counts from high UAV imagery were compared against both the fixed-wing imagery to derive updated detection and classification rates. A detailed description of methods is provided in Russell *et al.* [15].

All surveys (both UAV and fixed-wing) were counted with pups assigned as whitecoat, moulted or dead by a single experienced observer to ensure consistency. Detection and classification rates were estimated by comparing the classed counts, as well as matching individual pups across the two platforms. This matching was conducted by a different observer, and involved extensive scanning of the images to match pup and find pups missed on one the images. Detection rates for each platform were then estimated using the final count (i.e. accounting for any pups missed or not visible in either image). The final count overlaid onto the high-resolution UAV imagery was then used to divide the whitecoats into stages 1–3 versus stage 4.

Together, these data were used to derive new fixed detection and classification rates for objectives 4 and 5 of this study (see supplementary material, appendix S2, table S2.2). These rates were assumed representative of the remaining surveys in 2023 and were used to parameterize the observation process for all aerial surveys on Brownsman and Staple Islands in 2023, which were also counted by the same expert. Using these updated detection rates, we compared full model results from Brownsman and Staple Islands (2023) using both UAV and fixed-wing surveys with a model with fixed leaving process (TTL) using updated observation parameters to a model where TTL is estimated. See supplementary material, appendix S4, for sensitivity of pup production estimates to assumptions about TTL and interaction with the observation process.

Objective 5: Estimate moulting and leaving parameters from real data. We relaxed assumptions for TTM and TTL, allowing these transition parameters to be estimated from the data using moderately informative priors (table 1) that constrained estimates to biologically realistic ranges. We compared models with fixed versus estimated TTL using posterior predictive checks and Bayesian p -values to assess model fit. We tested these assumptions by fitting the model to the Brownsman and Staple Island data (both UAV

Table 3. Posterior mean estimates, standard deviations, 95% highest CRIs, relative percent bias and coverage for key parameters across 100 simulations from the model without mortality.

parameter	true value	mean estimate	s.d.	lower 95% CRI	upper 95% CRI	relative % bias	coverage
total births	500.00	523.53	15.59	492.13	555.26	4.71	0.69
mean birth date	34.00	33.25	0.61	32.02	34.50	−2.22	0.80
s.d. of birth date	12.00	15.39	1.04	13.32	17.51	28.24	0.09
skew	4.00	5.67	0.71	3.53	7.99	41.81	0.83

Table 4. Posterior mean estimates, standard deviations, 95% highest CRIs, relative percent bias and coverage for key parameters across 100 simulated datasets from the full model, which includes estimation of mortality rate.

parameter	true value	mean estimate	s.d.	lower 95% CRI	upper 95% CRI	relative % bias	coverage
total births	500.00	499.01	14.47	471.00	528.00	−0.20	0.96
mean birth date	34.00	34.21	0.70	32.80	35.60	0.62	0.95
s.d. of birth date	12.00	11.87	0.94	9.96	13.80	−1.08	0.95
skew	4.00	4.32	0.81	2.28	6.61	8.01	0.99

and fixed-wing surveys) using the updated observation parameters from objective 4 and evaluated both fixed and estimated TTM and TTL values. To test whether the results and fit to data were generalizable, we also fitted the model to the fixed-wing survey data from additional Farne Island colonies (see supplementary material, appendix S5).

3. Results

The MCMC results indicated convergence ($\hat{R} < 1.1$) for all model parameters and all models (across objectives).

3.1. Simulation studies

Objective 1: Mirroring standard model in Bayesian framework. Simulating a plausible level of pup mortality and omitting it from the model to mimic the standard model resulted in a positive bias for pup production estimates (+4.7%). Relative bias for the standard deviation of birth dates (σ_μ) was +28.2% and the skew parameter (α) had a relative bias of +41.8%. The 95% CRI coverage of true values was 0.69 for total births, 0.80 for mean birth date and 0.09 for the standard deviation (table 3).

Objective 2: Extend model to explicitly include mortality. Across all simulated datasets, the full model estimating mortality resulted in the expected 95% CRI coverage (95–99%) and low relative bias (−1.1 to +8.0%). The relative bias of the skew parameter in the full model was 8%; however, this represents a small absolute difference in the true simulated value of 4 and mean posterior estimate of 4.32 (table 4). Across 100 simulated datasets (that included mortality), pup production estimates in the full model where mortality was estimated showed negligible bias compared to the standard model that omitted pup mortality estimation. These results demonstrate that when mortality is present in the colony, explicitly modelling it in the estimation procedure reduces bias and improves inference at a mortality rate of 1.8% daily for newborn pups.

Objective 3: Sensitivity analyses. Sensitivity analyses demonstrated that pup production estimates were sensitive to misspecification of observation parameters and transition processes (figure 1). Misspecification of whitecoat detection probability (where the true simulated value is $\psi_w=0.95$) had the strongest effect on pup production estimates, with detection rates below 0.85 leading to substantial negative bias (greater than 10%). Moulded pup classification accuracy (κ_m) also influenced estimates, particularly when combined with misspecified TTL. Fixing κ_m at 0.50 (as in pre-2010 models) resulted in overestimation of

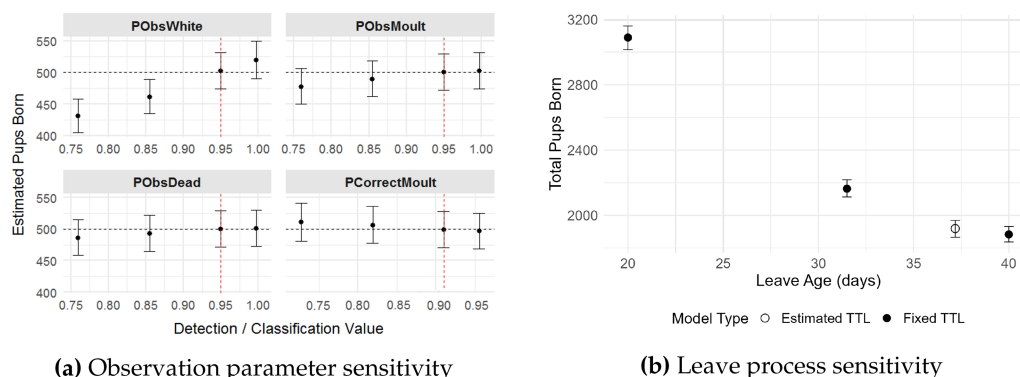


Figure 1. Key sensitivity analysis results. (a) Mean posterior estimates of total pups born across varying detection and classification probabilities (50 simulations, true number born = 500, vertical red dotted line indicates true parameter value). (b) Estimated pup production at Brownsman and Staple Island (2023) across a range of fixed TTL values, with model-estimated TTL shown for comparison. Detailed sensitivity results are provided in supplementary material, appendix S4.

Table 5. Full model posterior estimates of total pup production from UAV and fixed-wing surveys at Brownsman and Staple Islands (2023), fitted using previous and updated fixed-wing detection parameters (see supplementary material, appendix S2, for detailed methods and updated rates).

survey	detection/classification rates	median	95% CRI
fixed-wing	previous fixed-wing	2171	[2116, 2227]
	updated fixed-wing	2146	[2092, 2204]
UAV	previous fixed-wing	2111	[2075, 2149]
	UAV	2081	[2045, 2118]

pup production by 15–20% when TTL was simultaneously underestimated (see supplementary material, appendix S4).

Pup production estimates were relatively robust to moderate misspecification of TTM, with bias remaining below 5% for TTM values within 3 days of the true value. Contrastingly, misspecification of TTL had substantial effects on pup production. When the true leave age differed from the assumed value of 31.5 days, pup production was overestimated by 11–15% when the true leave age was 38 days and underestimated by approximately 26% when it was 22 days, across the range of leave-age standard deviations tested. These findings highlight the importance of accurate observation parameters and demonstrate that TTL is a critical process parameter requiring careful specification or estimation. Detailed results of all sensitivity analyses, including parameter interactions and figures showing bias across parameter ranges, are provided in supplementary material, appendix S4.

3.2. Real data

Objective 4: Survey platforms and updated observation parameters. Tabular comparisons from the model without mortality fitted to the same datasets can be found in supplementary material, appendix S2, table S2.3. Estimates derived from the five fixed-wing surveys produced slightly higher estimates of pup production at 2171 pups (95% CRI: 2116–2227; table 5), with a similar, but slightly narrower right-skewed distribution (95% CRI: 2.44–4.14).

For both survey platforms, explicitly modelling mortality and carcass decay consistently reduced estimated skew and variation in the birth curve, and slightly reduced overall births; figure 2 shows posterior median estimates from models with and without explicit modelling of pup mortality.

Objective 5: Estimate moulting and leaving parameters. Under the fixed TTL assumption and updated detection rates, the model fitted to UAV surveys estimated total pup production at 2081 pups (95% CRI: 2045–2118), while the model fitted to fixed-wing surveys estimated 2146 pups (95% CRI: 2092–2204). Allowing TTL to be estimated reduced total pup production to 1830 (95% CRI: 1793–1869) pups for

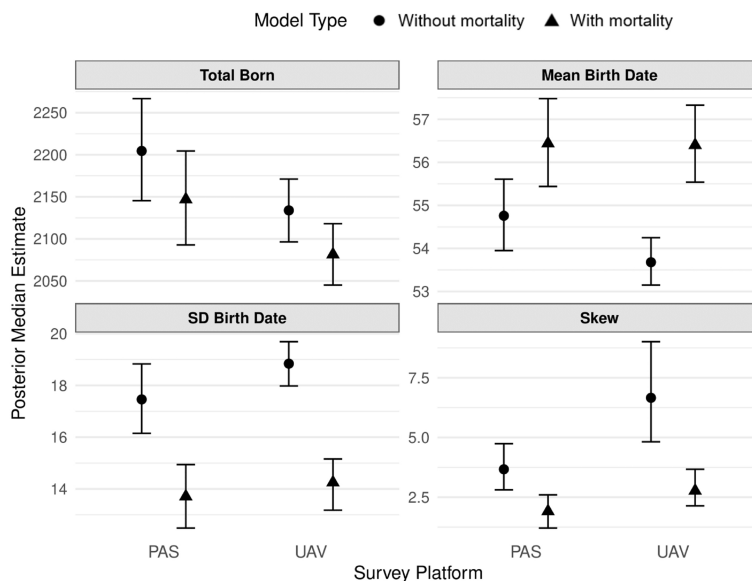


Figure 2. Posterior median estimates and 95% CRI for four key parameters (total births, shifted mean birth date and standard deviation of birth date (to account for skew and truncation), and skewness), shown separately for fixed-wing (PAS refers to the fixed-wing survey platform) and UAV surveys of Brownsman and Staple colonies (2023). Models either mimicked previous approaches that added dead pups to whitecoat counts ('Without mortality') or included mortality explicitly ('With mortality') and used updated observation parameters.

Table 6. Posterior estimates of key parameters with 95% CRI and medians for the UAV and fixed-wing datasets at Brownsman and Staple Islands (2023), using updated observation rates and allowing the model to estimate TTL and its standard deviation.

parameter	UAV survey		fixed-wing survey	
	median	95% CRI	median	95% CRI
total births	1830.33	[1793.36, 1868.59]	1905.05	[1847.60, 1960.47]
mean TTL	37.14	[36.55, 37.85]	38.00	[36.95, 38.93]
s.d. TTL	1.37	[1.00, 2.32]	1.39	[1.00, 2.67]

UAV survey estimates and 1905 (95% CRI: 1848–1960) for fixed-wing survey estimates, representing a reduction of 12.0% and 11.2%, respectively (table 6). TTL was estimated to be between 37 and 38 days of age across both survey platforms, with low standard deviation (1–3 days). See supplementary material, appendix S4, for sensitivity analysis of pup production estimates to assumptions about TTL and interaction with the observation process.

3.3. Posterior predictive checks: fixed-wing surveys

Posterior predictive performance of the model with fixed TTL and jointly estimated TTM and TTL is visualized in figure 3. The model showed poor fit according to Bayesian p -values when the leave process was fixed, particularly for moulted and dead pups. The fixed TTL model consistently overestimated dead counts (posterior predictive $p = 0.83$) and underestimated moulted pup counts ($p = 0.04$), while whitecoats were predicted with both over and underestimation across surveys ($p = 0.68$).

Allowing the model to estimate the leaving process (TTL and standard deviation of TTL) provided minor improvements to posterior predictive performance. Posterior predictive checks in the model where dead pups were not explicitly modelled as in previous models are given in supplementary material, appendix S5, fig. S5.7. Clear improvements were only observed when both moult and leave processes were estimated jointly, with TTM estimated at 20.2 days (95% CRI: 19.2–21.2), approximately 3 days earlier than the previously assumed fixed value of 23 days, with a similar standard deviation (s.d. = 7.1 days

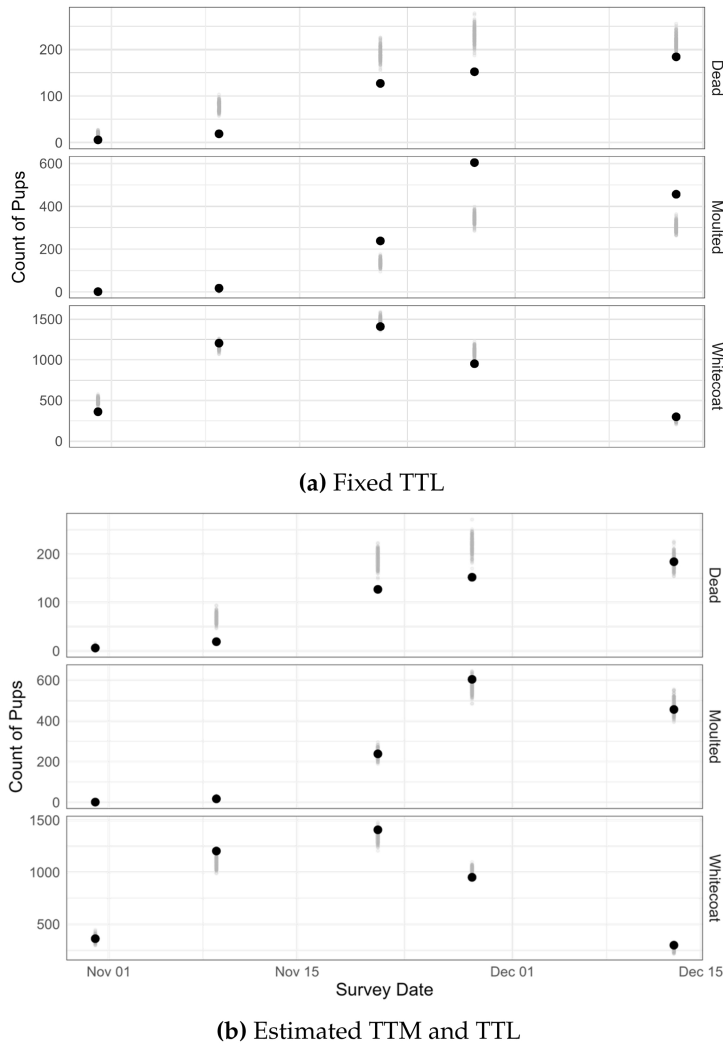


Figure 3. Posterior predictive checks for fixed-wing survey data under two model assumptions. Grey bars show posterior draws and black points are observed counts across pup stages (dead, moulted, whitecoat). Model fit improves when moult and leave processes are estimated. (a) Fixed TTL and (b) Estimated TTM and TTL.

(95% CRI: 5.9–8.4)) compared to the fixed value of 5 days. TTL was estimated at 35.9 days (95% CRI: 34.5–37.1) with s.d. = 2.3 (95% CRI: 1.1–5.2). This version of the model showed substantially improved fit for dead ($p = 0.69$) and moulted ($p = 0.47$) observations. For whitecoats ($p = 0.30$), the model tended to underpredict high observed counts. Under both model variants, dead pup counts were overpredicted in the middle surveys, suggesting the logit-linear survival function may not fully capture mid-season mortality patterns (see [S4](#)).

4. Discussion

We developed a flexible Bayesian framework to improve estimation of grey seal pup production and key life history parameters to address limitations in the standard maximum likelihood modelling approach. In line with objective 1, we successfully fitted the standard model in a Bayesian framework using the same fixed parameter values, number of pups, and survey scenario previous models have used to estimate pup production in the UK. We demonstrated that the Bayesian model can obtain results consistent with previous modelling approaches [14,16]. This provided a foundation on which to build further model extensions in a Bayesian framework. Building on this framework, objective 2 extended the model to include pup mortality and carcass persistence, quantified bias and uncertainty from simulated datasets, and provided results for the model fit to real data. Although little is known about the survival and

carcass persistence rates across the UK, this model lays the groundwork needed to explore the mortality process in more depth and conduct survival and carcass persistence sensitivity analyses. To address our third objective, we conducted sensitivity analyses to evaluate the sensitivity of pup production and the birth process to changes in observation and life-stage transition processes (see supplementary material, appendix S4). These sensitivity analyses were designed to identify which parameters most strongly influence pup production estimates. The results highlight that pup production estimates depend on accurate specification of detection parameters and TTL, motivating empirically derived detection rates in objective 4 and estimation of TTL in objective 5. We then addressed our fourth objective and fitted the model using updated observation rates from newly available data and compared results from previous observation process parameters. Finally, we addressed our fifth objective and extended the model to estimate the moult and leaving parameters.

We made two modelling choices to balance parsimony, computational efficiency and biologically realistic processes. First, to improve computational efficiency, we used an interval-based birth process rather than daily births. Simulations (see supplementary material, appendix S3), demonstrated that the interval-based model introduced a moderate negative bias in pup production estimates (4.9%) while substantially reducing computation time. Although interval births are sufficient for the objectives of this study, a daily birth process would reduce bias, increase the fine-scale temporal resolution and could be useful for future work examining UK pup production estimates. The reduced coverage at $\tau = 4$ (70%) is due to the interval discretization approximation. Coverage is 93% when births are both simulated and fitted at daily resolution. The tau sensitivity analysis demonstrates that coverage degrades with interval width, and CRIs for pup production under $\tau = 4$ are narrower than $\tau = 1$ and should be interpreted with this in mind. Second, we used a Poisson likelihood rather than an MVN approximation to the multinomial. Simulations showed that these likelihoods produced near-equivalent estimates for total pup production and similar CRI widths (see supplementary material, appendix S3). The Poisson approximation was retained for this study owing to its parsimony. Future applications of this model to estimate pup production across the UK and through time may benefit from revisiting these choices as computational time and resources allow. In addition, simulations in this study were conducted under the correctly specified model. Future work could assess model performance under misspecification, including alternative birth distributions or transition dynamics, to provide a more thorough evaluation of robustness.

We extended the model to explicitly represent pup mortality and compared results to the standard model (objective 2). When the standard model (omitting survival and carcass detection) was fitted to simulated data, in which dead pups were present but added to the whitecoat counts, moderate positive bias was introduced to pup production and substantial bias in estimates of breeding phenology parameters. Under the standard model, mortalities are added to the live whitecoat counts. Our results show that explicitly modelling mortality reduced the estimated standard deviation and skew of the birth curve, increased estimated mean birth date (by 1–2 days) and reduced estimated total births. This slight shift to later birth date may indicate that the model uses information about carcass persistence to narrow the birth distribution and shift the mean. By accounting for mortality, the model can attribute dead pups persisting later in the season as earlier births rather than prolonging the breeding season and extending the right tail of the distribution. This suggests that it is critical to model mortality when the goal is to accurately characterize the timing, shape and spread of the breeding season, especially if future research aims to make inference across years or colonies. Pup mortality varies across the season and spatially [24–26]. Carcass persistence also varies spatially, with carcasses closer to shore being more likely to wash away and be unavailable to count than those more interior to the colony. Scavenging may also influence the duration a carcass remains available to count. Future work could leverage empirical carcass data (tracking known carcass locations across surveys) to parameterize this aspect of the mortality process, potentially allowing carcass persistence to vary by age. Our framework lays the foundation to explicitly model survival and carcass persistence, but reliable inference will depend on careful parameterization and empirical data. In addition, a more flexible survival function, such as incorporating a quadratic term ($\text{logit}(p) = \beta_0 + \beta_1 a + \beta_2 a^2$), may improve fit to dead pup counts in the middle of the season, though limited empirical data on age-specific survival rates currently limits the ability to inform this parameter reliably.

We incorporated updated detection and misclassification rates in both the standard and full model using rates derived from the rare tandem UAV and fixed-wing survey data at Brownsman and Staple Island in the Farne Islands, UK. Updating these observation parameters led to modest reductions in pup production for both survey platforms indicating that the standard model may overestimate pup

production. The sensitivity analysis in supplementary material, appendix S4, demonstrates the importance of accurate specification of detection rates within plausible biological ranges to ensure unbiased inference of pup production.

Two key process parameters influence pup production estimation: moult (TTM) and leaving (TTL). We extended the model and allowed TTM and TTL to be estimated. Compared to the mirrored maximum likelihood model (objective 1) with fixed parameter values, the flexible Bayesian model demonstrated improved fit to real data at Brownsman and Staple Island (and to a second dataset: Inner and Outer Farne Islands for 2023; supplementary material, appendix S5). Ultimately, sensitivity analyses demonstrated pup production is fairly robust to moderate changes in TTM (see supplementary material, appendix S4, fig. S4.6). TTL controls how long a pup is expected to stay on the colony such that an extended TTL increases the number of individuals on the colony across surveys. This means that as TTL increases, the model will generate fewer new births to explain the observed data and total pup production will be reduced. Allowing the model to estimate TTL led to reductions in pup production estimates compared to the fixed model. This result supports previous studies [27,28] indicating the currently used value for TTL of 31.5 is too low. In the UK, the value of 31.5 days is based on a previous study that tracked pups from birth until they left the colony for sea; however, some pups were still on the colony when the study concluded [10], and this probably negatively biased the estimate of TTL.

The sensitivity analysis in supplementary material, appendix S4, highlights the importance of accurately informing the priors on these parameters to ensure a biologically plausible range when TTL is being estimated. However, there are other factors that complicate reliably estimating TTL at a particular colony; Russell *et al.* [14] suggest TTL may vary with habitat, weather and colony density and that it is possible pups born later in the season do not stay as long on the colony. To account for the variability and uncertainty in these processes within and among colonies, future model development could incorporate variation in the leaving process, allowing TTL and its standard deviation to vary throughout the season and potentially across colonies.

One key future model extension will be building a hierarchical structure that enables the model to capture the spatiotemporal variation by treating birth and life-stage transitions as hyperparameters and allowing for spatial and where applicable, temporal variation of these parameters. Realistically, several key biological processes, particularly birth timing, are variable at a colony and year level. By enabling the sharing of information from data-rich colonies and years to inform estimates for data-poor colonies in a hierarchical framework, future studies can test whether this provides enough statistical power to locally or regionally estimate the birth process, leaving and survival processes, and if this partial-pooling framework can reduce overall survey effort. This model modification could reduce bias in pup production estimates and potentially resolve identifiability issues in some of the observation processes. Finally, a hierarchical framework would allow for investigation into changes in breeding phenology and potential environmental drivers of change in a spatially structured way.

Ethics. Ethical approval for the aerial surveys and use of the resulting data was granted by the University of St Andrews (approval codes: SEC21032; BL17766).

Data accessibility. Stan model code and empirical data are available [29].

Supplementary material is available online [30].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. M.R.G.: conceptualization, formal analysis, investigation, methodology, software, validation, visualization, writing—original draft, writing—review and editing; K.T.S.: formal analysis, methodology, software, supervision, writing—review and editing; E.K.J.: conceptualization, funding acquisition, methodology, supervision, writing—review and editing; L.T.: conceptualization, funding acquisition, methodology, supervision, writing—review and editing; C.D.M.: data curation, investigation, resources; S.A.W.: data curation, investigation; C.D.D.: investigation; D.J.F.R.: conceptualization, data curation, funding acquisition, methodology, project administration, resources, supervision, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

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